Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

EARTH TESTER · HAND-DRIVEN · METAL BODY

CIE/222M Range Guide

3-terminal and 4-terminal, any range 0–1 Ω to 0–10,000 Ω

Single range or multi range, 3 or 4 terminal

The CIE/222M measures earth resistance (3-terminal) or earth resistance plus soil resistivity (4-terminal). Choose the terminal type, then the range configuration. Every range below is available in both terminal types.

3 / 4	1–5	10,000 Ω
TERMINAL TYPES	RANGES PER INSTRUMENT	HIGHEST RANGE



CIE/222M · ILLUSTRATIVE IMAGE

01 SINGLE RANGE

RANGE BAND	SELECTION
0–1 Ω to 0–5 Ω	Any range within this band
0–10 Ω to 0–100 Ω	Any range within this band
0–200 Ω to 0–10,000 Ω	Any range within this band

Any single range from 0–1 Ω up to 0–10,000 Ω can be specified within the three bands above.

02 DUAL RANGE

CONFIGURATION	FULL-SCALE RANGES (Ω)
Dual range	0–1 / 10 · 0–2 / 10 · 0–3 / 30 · 0–10 / 100

03 3-TERMINAL AND 4-TERMINAL

3-terminal type

Measures earth resistance. Connect the earth electrode under test and two test spikes.

4-terminal type

Measures earth resistance and specific soil resistivity, using four spikes in a line.

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Multi-range options, specifications, kits and ordering

04 TRIPLE, FOUR AND FIVE RANGE

CONFIGURATION	FULL-SCALE RANGES (Ω)
Triple range	0–1 / 10 / 100 · 0–2 / 10 / 100 · 0–2 / 20 / 200 · 0–3 / 30 / 300 · 0–10 / 100 / 1,000
Four range	0–1 / 10 / 100 / 1,000 · 0–2 / 10 / 100 / 1,000 · 0–10 / 100 / 1,000 / 10,000
Five range	0–1 / 10 / 100 / 1,000 / 10,000 · 0–2 / 10 / 100 / 1,000 / 10,000

Multi-range instruments carry a range selector switch; the ranges are read on one scale. Any other range can be manufactured against specific requirements.

05 KEY SPECIFICATIONS

Body	Anti-magnetic aluminium
Generator Type	Hand-driven generator with centrifugal governor
Scale Length	80 mm (approx.)
Accuracy	±5% (as per IS: 9223/1979)
Generator Rotation Speed	160 r.p.m.
Weight	2.5 kg (approx.)
Dimensions (L×W×H)	110 × 128 × 190 mm (approx.)

06 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Testing kit, 3-terminal: 2 M.S. spikes (45 cm), 3 flexible-wire spools (10 / 20 / 30 m), hammer, carrying bag
- Testing kit, 4-terminal: 4 M.S. spikes (45 cm), 4 flexible-wire spools (10 / 20 / 20 / 30 m), hammer, carrying bag
- Carrying case for earth tester

07 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example `CIE/222M, 4-terminal, 0–1 / 10 / 100 Ω`. Any range not listed can be manufactured against specific requirements - enquire.

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